VSFA: Visual and Scattering Topological Feature Fusion and Alignment Network for Unsupervised Domain Adaptation in SAR Target Recognition

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***Abstract*— In recent years, how to accurately identify targets from measured synthetic aperture radar (SAR) images according to electromagnetic synthetic SAR images is attracting more and more research interest. Most existing algorithms only focus on decreasing the domain differences in visual representations, while the special characteristics of SAR images are not explored enough. Besides, these algorithms tend to align only the overall distribution of synthetic and measured images, while domain shifts between subclasses are ignored. To solve these problems, a novel unsupervised domain adaptation framework named visual and scattering topological feature fusion and alignment network (VSFA) is proposed in this article. First, considering that visual features are crucial for recognition, image reconstruction is introduced to enhance the generalization of visual features. Second, by analyzing the imaging mechanism of SAR, we explore the differences in scattering topologies between synthetic and measured images for the first time. To measure the differences quantitatively, we model the non-Euclidean scattering topology of the target as graph data and introduce graph neural networks (GNNs) to extract scattering topological features. Moreover, in order to describe the scattering topology of the target more comprehensively, we introduce two different but complementary scattering topological point extraction algorithms and achieve their fusion at the feature level by GNN for the first time. Finally, a simple but effective two-stage domain adaptation loss is proposed to constrain the network to align the distribution of synthetic and measured images class by class. Benefiting from the simultaneous reduction of distribution differences in visual space and scattering topological space, the proposed method achieves 99.15% and 98.18% recognition accuracies in two typical experiment scenarios of the Synthetic and Measured Paired Labeled Experiment (SAMPLE) dataset, demonstrating its effectiveness.**

***Index Terms*— Automatic target recognition (ATR), convolutional neural network (CNN), graph neural network (GNN), scattering topology, synthetic aperture radar (SAR), synthetic data.**

I. Introduction

BENEFITING from its unique electromagnetic scattering imaging mechanism, synthetic aperture radar (SAR) can operate under all-day, all-weather, and long-range conditions, which makes SAR widely used in military and civilian fields. Compared with optical images, SAR images lack color information and are susceptible to interference from speckle noise, which makes interpreting SAR images more difficult. Automatic target recognition (ATR) is a critical topic in the research of intelligent interpretation of SAR images and has received increasing attention from researchers.

Most early traditional recognition algorithms relied on well-designed handcrafted features, which required expert knowledge and constrained the development of recognition algorithms. With the rapid development of deep learning in recent years, researchers proposed many recognition algorithms based on convolutional neural networks (CNNs) [1], [2], [3]. Thanks to the powerful feature extraction and combination capability, the performances of CNN-based algorithms significantly outperform traditional algorithms. However, deep learning is a data-driven method, which means that its high recognition accuracy is built on a large amount of training data. Unfortunately, obtaining and labeling sufficient SAR images are time-consuming and expensive. Especially for noncooperative targets, it is not practical to obtain their measured images. The lack of samples brings great challenges to the design of deep learning-based SAR ATR algorithms.

Data augmentation is a mainstream method that can alleviate data insufficiency. Simple image transformations, such as cropping, rotating, and adding noise, can improve the recognition performance to some extent [1], [2] but are still constrained by the size of the original dataset. With the development of generative models, some researchers generate SAR images by autoencoders [4] and generative adversarial networks (GANs) [5], [6]. However, the generated images often lack local details, so it is difficult to guarantee their authenticity. Considering the unique imaging mechanism of SAR, i.e., imaging based on the backscattered echoes of targets, other researchers generate synthetic SAR images based on the computer-aided design (CAD) models of targets and electromagnetic simulation techniques [7], [8], [9], [10]. Generally speaking, the synthetic images produced by the last method are more reliable and physically meaningful.

However, synthetic images cannot completely replace measured images. In the simulation program, the CAD model, the ground background, and the propagation process of electromagnetic waves do not precisely match the

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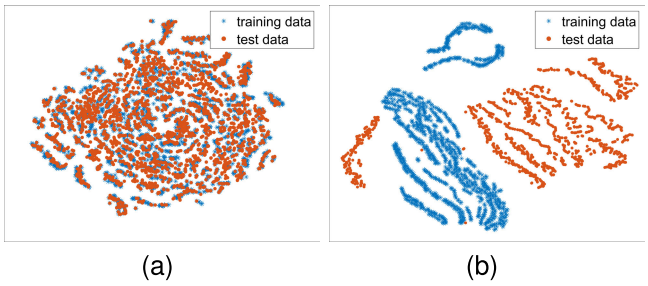


Fig. 1. Data distribution visualization based on t-SNE embeddings [12]. (a) MSTAR dataset [11]. (b) SAMPLE dataset [7]. The MSTAR dataset is composed entirely of measured data, while the training set of the SAMPLE dataset is synthetic data.

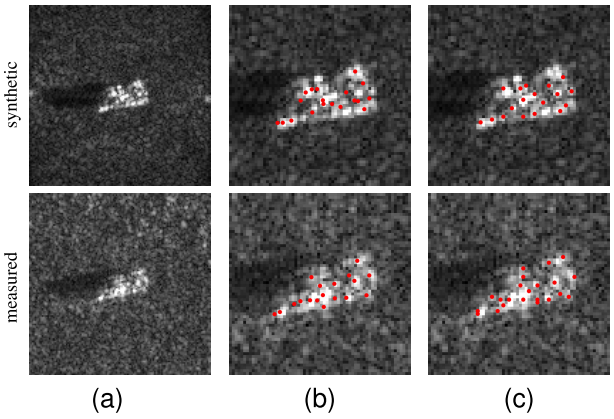


Fig. 2. Differences between the synthetic and measured images. (a) Magnitude images. (b) Scattering topological points extracted by the ASC model [20]. (c) Scattering topological points extracted by the SAR-SIFT algorithm [21]. To show the distribution of scattering topological points more clearly, we crop the original image and resize the target area in (b) and (c). Each red point denotes one scattering topological point.

actual physical situation, which leads to domain distribution differences between synthetic SAR images and measured SAR images. In other words, the synthetic and measured images do not satisfy the assumption of independent and identical distribution. Therefore, the classifier trained on synthetic images cannot generalize well to measured images. To facilitate understanding, the distributions of training data and test data in the Moving and Stationary Target Acquisition and Recognition (MSTAR) dataset [11] and the Synthetic and Measured Paired Labeled Experiment (SAMPLE) dataset [7] are visualized in Fig. 1. Specifically, the t-distributed stochastic neighbor embedding (t-SNE) [12] is used to embed high-dimensional images into a 2-D plane. Each point represents one image. Unlike the MSTAR dataset, which consists entirely of measured images, the training set of the SAMPLE dataset is generated by the electromagnetic simulation program. As shown in Fig. 1, the training and test data in the MSTAR dataset satisfy the assumption of independent and identical distribution. At the same time, there is a significant distribution difference between the training and test data in the SAMPLE dataset. Therefore, the recognition decision boundaries trained by synthetic images are not applicable to measured images [7], [8], [9], [10]. In recent years, how to bridge the distribution gap and accurately identify measured images by synthetic images has become a research hotspot.

At present, some scholars have studied this problem. Early researchers focused on improving recognition performance with a large number of synthetic images when few labeled measured images were available [13], [14], [15], which is a relatively simple assumption. More recently, researchers are interested in the more challenging condition where no labeled measured images are available during the training process. Inkawhich et al. [16] explore how to improve recognition performance from four aspects: data preprocessing, model construction, loss function choices, and ensemble learning. However, the effectiveness of their algorithm largely derives from adding noise to synthetic images, so their algorithm is strongly data-dependent. Zhang et al. [8] propose a two-stage recognition algorithm that uses the CNN to identify easy samples and the multisimilarity fusion classifier to identify hard samples. Nevertheless, the multisimilarity fusion classifier is based on template matching, which requires a lot of template images and is computationally inefficient. In addition, some researchers have proposed several recognition algorithms [9], [10], [18] based on the domain adaptation method [19] and achieved exciting results. Due to the ability to adaptively extract domain-invariant features and reduce the distribution differences between different data in the feature space, the domain adaptation method has more potential for cross-domain recognition tasks. Thus, our proposed algorithm is also based on the domain adaptation method. In the domain adaptation method, the domain with rich supervisory information is often named the source domain, while the domain with little or no labeled samples is named the target domain. It is obvious that, in this article, the synthetic data are the source domain, and the measured data are the target domain.

However, there are two problems in the above domain adaptation algorithms. The first problem is that they all feed magnitude SAR images directly into CNNs to extract visual features, which essentially treats SAR images as optical images and ignores the imaging differences between SAR images and optical images. Unlike optical cameras, SAR produces images based on the backscattered echoes of targets. As shown in Fig. 2(a), since different local structures have different backscattered intensities for electromagnetic waves, the imaging results of different parts of a vehicle target often vary in brightness. For example, dihedral and trihedral structures in vehicles often form strong scattered echoes, and thus, these areas appear brighter in the image. Besides, it can be found that the overall distribution of these local bright regions is able to reflect the specific scattering topology of the target. In this article, we use points to represent these regions and name them as the scattering topological points of the target. Fig. 2(b) and (c) shows the scattering topological points extracted by two different algorithms, i.e., the attribute scattering center (ASC) model [20] and the SAR-scale-invariant feature transform (SAR-SIFT) algorithm [21]. Due to the inaccurate simulation process, for targets of the same category in the synthetic and measured images, in addition to visual differences, there are some differences in the number and position of scattering topological points, which leads to different scattering topologies. Therefore, reducing the distribution differences between the scattering topologies

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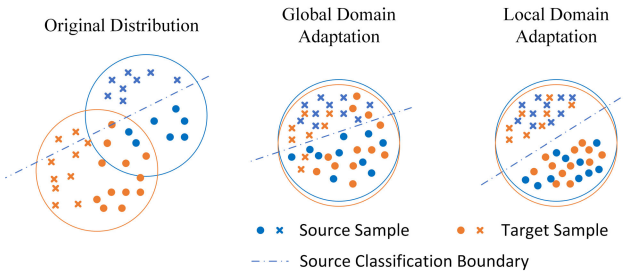


Fig. 3. Different domain adaptation method.

of the synthetic and measured images in the feature space should be helpful for the cross-domain target recognition task. However, it is quite difficult to model scattering topologies and quantitatively measure the difference between different scattering topologies due to their non-Euclidean properties.

The second problem is that they tend to align only the overall distribution between the source and target domains while ignoring the differences between subdomains. As shown in Fig. 3, when the global distribution between the source and target domains has been aligned, domain shifts may still exist between their subdomains, resulting in some samples being classified in the wrong category. Recently, some researchers proposed a local domain adaptation loss [22] that assigns the target domain samples into different classes according to their prediction confidences and then aligns the target domain samples and the source domain samples class by class. Because of alleviating the problem of subdomain shift, its performance significantly outperforms global domain adaptation methods [23], [24]. However, the computation process of local domain adaptation loss brings a new problem. Misclassified samples will participate in the subdomain alignment of noncorresponding classes with high weights, resulting in inaccurate subdomain alignment. This negative impact will grow deeper as the training progresses.

To address the above problems, a novel unsupervised domain adaptation framework for SAR ATR named visual and scattering topological feature fusion and alignment network (VSFA) is proposed in this article. Unlike previous algorithms, the proposed algorithm takes into account both visual and scattering topological differences between the synthetic and measured images and aligns their distributions in the feature space class by class. First, visual information is usually crucial for recognition tasks. To extract more generalized visual features, we design a convolutional autoencoder [25] and introduce image reconstruction as an auxiliary task. Second, we introduce graph structures that can naturally represent non-Euclidean data to model the scattering topology of the target. In particular, two different algorithms are utilized to extract the scattering topological points of the target, which describes its scattering topology from different aspects and provides complementary information. Then, graph neural networks (GNNs) [26] designed for graph data are introduced to learn the structural information of the target and extract scattering topological features. Finally, to reduce the negative effect of misclassified samples on subdomain alignment, we combine the global domain adaptation loss and the local domain adaptation loss to propose a two-stage loss that can align the distribution of the synthetic and measured images more accurately. Instead of measuring visual differences and scattering topological differences between the synthetic and measured images separately, visual features and scattering topological features are fused and transformed into a more robust and efficient feature. In addition, VSFA employs a strategy of jointly training CNN and GNN, allowing the network to adaptively determine the contribution of each component to the recognition task. The satisfactory results of ablation experiments and comparison experiments demonstrate the effectiveness of the proposed method.

The main contributions in this article are summarized as follows.

1) A novel unsupervised domain adaptation framework named VSFA is proposed in this article. It simultaneously reduces visual differences and scattering topological differences between measured and synthetic SAR images for the first time and achieves state-of-the-art performance on the public SAMPLE dataset.
2) This article utilizes GNNs to aggregate and fuse the ASC features and SAR-SIFT features of the target for the first time, thus describing the local scattering characteristics and spatial topology of the target more comprehensively.
3) A simple but effective two-stage domain adaptation loss is proposed in VSFA. The introduction of the global domain adaptation loss significantly reduces the classification confidence of misclassified samples, which helps the local domain adaptation loss to align different subdomains more accurately in feature space.

The remainder of this article is organized as follows. The related works are briefly reviewed in Section II. Section III presents the details of our proposed method. Section IV provides the experimental results and the corresponding analyses. Finally, Section V concludes this article.

II. Related Works

A. Domain Adaptation and Related Algorithms

Most modern target recognition algorithms rely on large amounts of labeled data to achieve high accuracy. However, collecting and labeling massive amounts of data for each new task in many application scenarios are impractical. Therefore, it is necessary to transfer knowledge from existing similar tasks (source domain) to the new task (target domain). Unfortunately, there are some distribution differences between the data of similar domains, which limits knowledge transfer. Domain adaptation is a machine learning method that can bridge the distribution gap and effectively transfer knowledge between similar but different domains [22], [23], [24]. Unsupervised domain adaptation is applicable to the case where the target domain is unlabeled. The core of unsupervised domain adaptation is to extract domain-invariant features and reduce the distribution differences between the source and target domain data in the feature space.

At present, some researchers have applied domain adaptation methods to recognition tasks based on synthetic images. He et al. [9] propose a task-driven domain adaptation algorithm, which can mitigate the sensitivity of SAR images

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to depression angles by reducing the distribution difference of the synthetic data at different depression angles. Du et al. [10] propose an adversarial coding network combining reconstruction and domain adversarial training to transfer the physical features contained in the synthetic data to the measured data as a supplement. Wang et al. [17] propose an algorithm combining metalearning and domain adversarial training to achieve cross-domain and cross-task transfer learning by pretraining the network with a large amount of synthetic data and then fine-tuning the network using a small amount of measured data. Sun et al. [18] design a multiscale feature extraction module to extract more generalized features and improve performance through ensemble learning. Sun et al. [61] introduce the pseudolabel method [62] in unsupervised domain adaptation and propose a gradual domain adaptation recognition framework with a pseudolabel denoising method. The above studies have shown the prospect of applying domain adaptation methods to synthetic data. However, they all focus only on the global visual differences while ignoring the special imaging characteristics of SAR images. Moreover, except for [61], all other methods align only the whole distributions of the synthetic and measured images, ignoring subdomain shifts.

B. Scattering Topological Point and Related Algorithms

Due to the unique electromagnetic scattering imaging mechanism, SAR images have different characteristics from optical images. As shown in Fig. 2, the scattering intensity of different areas in the target is different. In other words, the imaging result of the target is composed of many scattering topological points. These scattering topological points can not only reflect the local scattering characteristics of the target but also represent the structure information of the target. Therefore, scattering topological points are widely and relatively successfully used in SAR target recognition [8], [27], [28], [29], [30], [31] [32]. These algorithms can be divided into two steps: scattering topological point extraction and classification. The extraction process and the classification process both vary in different algorithms. To extract scattering topological points, Ding et al. [27], Zhang et al. [28], [30], Feng et al. [29], and Li et al. [31] utilize the ASC model, while Zhang et al. [8] and Sun et al. [32] introduce different keypoint detection algorithms.

Generally speaking, the scattering topological point set extracted from the image is often non-Euclidean data, that is, there is no order between scattering topological points, and the number of neighbor scattering topological points of each scattering topological point is not consistent, which brings great difficulties to the design of recognition algorithm. Zhang et al. [8] and Ding et al. [27] use template matching algorithms to identify the targets, which requires a large number of template images and is time-consuming. Specifically, the former utilize the Hausdorff distance [33] to match the test image with the template image, while the latter use the Hungarian algorithm [34]. The algorithms proposed in [28], [29], [30], and [31] all postprocess the extracted scattering topological points to turn them into Euclidean data. Zhang et al. [28] introduce the bag of visual words method and generate a fixed size feature for each scattering topological point set. Feng et al. [29], Zhang et al. [30], and Li et al. [31] all reconstruct the extracted scattering topological points into an image and then send it to a CNN for classification. However, the postprocessing usually loses the original information of scattering topological points. Sun et al. [32] design an auxiliary task to predict the location of scattering topological points while training the classifier, which only utilizes the location information. The above algorithms either have a large amount of computation or lose the original information contained in scattering topological points. In addition, since the number and location of scattering topological points extracted from the same image by the ASC model and the keypoint detection algorithm are often different, it is difficult to utilize these two kinds of methods simultaneously.

C. Graph Neural Network and Related Algorithms

Graph data have attracted increasing attention from researchers in recent years due to their ability to naturally express numerous non-Euclidean data in the real world, such as social networks [35] and biochemical structures [36]. However, the complexity of graph data has imposed significant challenges on the existing machine learning algorithms. For example, the standard convolution cannot be used on graph data since it is not translation invariant. Recently, some researchers have successfully extended the convolution operator to graph data and proposed different GNNs. GNNs fall into two main streams: spectral-based [37], [38], [39] and spatial-based [40], [41], [42]. The early spectral-based approaches define graph convolution operators from the perspective of spectral graph theory and graph signal processing, which are poorly generalized despite their strong theoretical nature. The spatial-based approaches define graph convolution operators by message passing mechanism [41] and play a dominant role in current GNNs due to their flexibility and generality.

At present, there are few algorithms that apply GNNs in the SAR field. Li et al. [43] and Yang et al. [44] combine GNN and metalearning for few-shot SAR target classification. They regard the whole SAR images as nodes and introduce GNNs to learn the relationship between different images. In fact, they still do not take advantage of the topology information of the target. Li et al. [45] utilize GNN to learn the structure information of the target from its local features. However, they do not fully exploit the local scattering information of the target and completely ignore the visual information.

III. Methodology

To facilitate understanding, our proposed VSFA is visualized in Fig. 4, which comprises four parts: the visual feature extraction module, the scattering topological point extraction module, the scattering topological feature extraction module, and the feature fusion and classification module. As shown in Fig. 4, the network is completely parameter sharing for source and target domains, and the feature extraction process is also the same. A lightweight convolutional autoencoder is designed in the visual feature extraction module to extract

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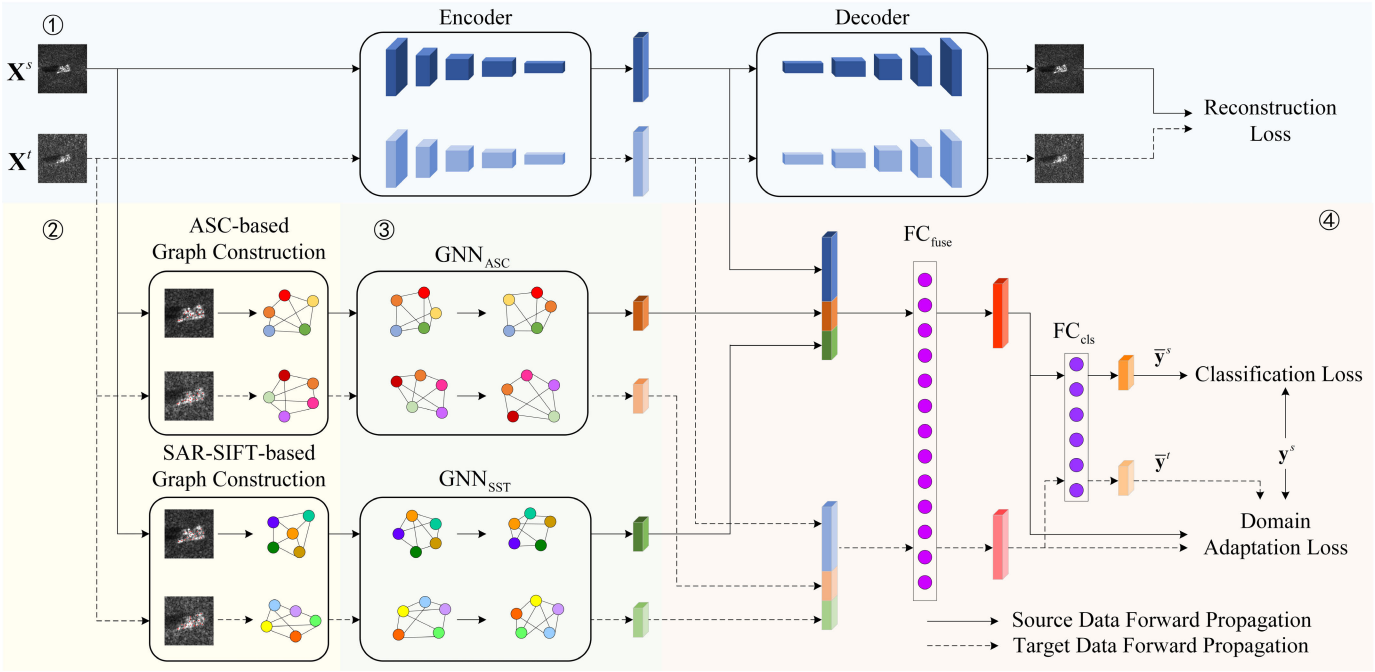


Fig. 4. Overall structure of the proposed VSFA. Note that the network is completely parameter sharing for both source and target domains. The feature extraction process is also the same. The differences appear only in loss calculations. The source domain is involved in the calculation of classification loss, while the target domain is not. Moreover, in addition to the features of the source and target domains, source domain true labels $\mathbf{y}^s$ and target domain predicted labels $\bar{\mathbf{y}}^t$ are also used to calculate the domain adaptation loss, which will be explained in detail in Section III-E. For simplicity, the subscript $(\cdot)_{SST}$ is used to abbreviate the SAR-SIFT algorithm.

visual features with better generalization capability. In the scattering topological point extraction module, the scattering topological points of the target are extracted from the image by the ASC model and the SAR-SIFT algorithm, respectively. Furthermore, graphs are constructed to model the overall distributions of scattering topological points. After transforming images into graphs, two GNNs are designed to extract scattering topological features from ASC-based graphs and SAR-SIFT-based graphs, respectively. In the feature fusion and classification module, visual features and two different but complementary scattering topological features are fused and delivered to fully connected (FC) layers to predict the labels. In particular, the proposed two-stage domain adaptation loss is calculated on the features output from the first FC layer. In this section, the four modules in Fig. 4 will be described in detail. For simplicity, the classification loss is denoted as Loss_{cls}, the reconstruction loss is denoted as Loss_{rec}, and the domain adaptation loss is denoted as Loss_{da}.

A. Problem Formulation

Suppose that we are given source domain labeled data $\mathcal{D}^s = \{(\mathbf{X}_i^s, \mathbf{y}_i^s)\}_{i=1}^{n^s}$ of n^s labeled samples ($\mathbf{y}_i^s \in \mathbb{R}^L$ is a one-hot vector indicating the label of $\mathbf{X}_i^s$ where L is the number of classes) and target domain data $\mathcal{D}^t = \{\mathbf{X}_j^t\}_{j=1}^{n^t}$ of n^t unlabeled samples. The goal of domain adaptation is to learn a model to reduce the distribution differences and minimize the prediction error $\sum_{j=1}^{n^t} \|\bar{\mathbf{y}}_j^t - \mathbf{y}_j^t\|$ in the target domain data, where $\bar{\mathbf{y}}_j^t$ is the predicted label of the target domain sample $\mathbf{X}_j^t$, and $\mathbf{y}_j^t$ is the corresponding true label that is unknown in the training process.

B. Visual Feature Extraction Module

For SAR image classification tasks, visual features are crucial and cannot be ignored. In recent years, CNNs have become the mainstream method for visual feature extraction due to the powerful capability of feature extraction and combination. Many excellent network architectures have been proposed, such as ResNet [46] and EfficientNet [47]. However, these networks are proposed for large-scale optical datasets and, thus, often have a large number of parameters to learn. Compared with optical datasets, SAR image datasets usually have the characteristics of fewer samples and smaller image sizes. Therefore, we design a lightweight CNN to extract visual features. Since Wang and Deng [19] state that simultaneous reconstruction of images from different domains allows the network to learn some domain-invariant representations through the shared encoder and maintain the domain-specific representations through the reconstruction loss, this network is designed as an encoder–decoder architecture [25], and image reconstruction is introduced as the auxiliary task.

The overall CNN architecture is shown in Fig. 5. It can be found that the encoder branch consists of only five convolutional layers with 32, 64, 128, 256, and 512 convolutional kernels, while the decoder branch consists of only five transposed convolution layers with 256, 128, 64, 32, and 1 convolutional kernels. *ks* and *st* denote the kernel size and the stride, respectively. Instead of using the pooling layer, the convolution operation with a stride size of 2 is used for downsampling. Except for the last transposed convolution layer that uses the hyperbolic tangent (Tanh) activation function [48], the rectified linear unit (ReLU) activation function [48] is applied

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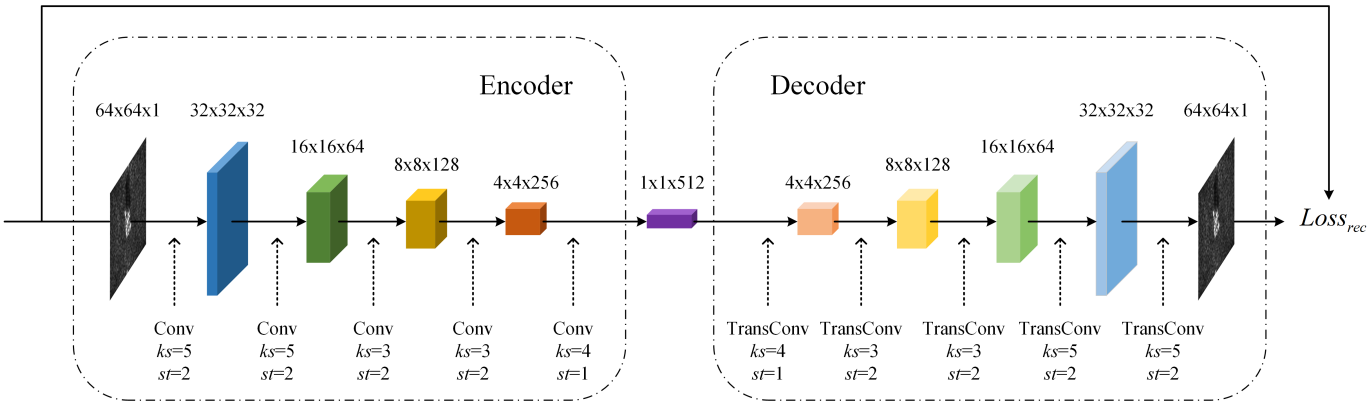


Fig. 5. Overall architecture of the CNN designed in this article. *ks* and *st* denote the kernel size and the stride, respectively.

after every convolution layer or transposed convolution layer, which is omitted in Fig. 5 for concision. In order to highlight the target region and reduce the computation cost, original images are center cropped into 64×64.

Mathematically, for any sample $\mathbf{X}$, the encoder branch can be parameterized as

$$\mathbf{z}_{\text{vis}} = \text{Encoder}(\mathbf{X}; \boldsymbol{\omega}_{\text{en}}) \tag{1}$$

where $\boldsymbol{\omega}_{\text{en}}$ and Encoder$(\cdot)$ denote the parameters and the nonlinear mapping process of the encoder branch, respectively. $\mathbf{z}_{\text{vis}}$ represents the extracted visual features. Similarly, the decoder branch can be parameterized as

$$\hat{\mathbf{X}} = \text{Decoder}(\mathbf{z}_{\text{vis}}; \boldsymbol{\omega}_{\text{de}}) \tag{2}$$

where $\boldsymbol{\omega}_{\text{de}}$ and Decoder$(\cdot)$ denote the parameters and the nonlinear mapping process of the decoder branch, respectively. $\hat{\mathbf{X}}$ is the reconstructed image. The mean square error (MSE) loss is used in this article as the reconstruction loss to constrain the realism of the reconstructed images. For all source samples and target samples, the reconstruction loss can be defined as

$$\text{Loss}_{\text{rec}} = \frac{1}{2}\left(\frac{1}{n^s}\sum_{i=1}^{n^s}\left\|\hat{\mathbf{X}}_i^s - \mathbf{X}_i^s\right\|_F^2 + \frac{1}{n^t}\sum_{j=1}^{n^t}\left\|\hat{\mathbf{X}}_j^t - \mathbf{X}_j^t\right\|_F^2\right) \tag{3}$$

where $\|\cdot\|_F^2$ denotes the squared matrix Frobenius norm.

C. Scattering Topological Point Extraction Module

As mentioned in Section I, scattering topological points can provide abundant information about the target, which is beneficial for classification. To more comprehensively explore the local scattering characteristics of the target, we introduce two different scattering topological point extraction algorithms, namely, the ASC model and the SAR-SIFT algorithm. These two algorithms have both similarities and differences. The similarities are that, although the scattering topological points extracted from the same image by both algorithms may differ in exact number and specific location, their overall distributions are highly similar, which can be found in Fig. 2. This phenomenon is reasonable and indicates that both algorithms are able to describe the topology of the target well. The differences lie in the features corresponding to scattering topological points. The former mainly reflect physical attributes such as magnitude, position, shape, and orientation of scattering centers, while the latter mainly reflect edge information through gradient. Different features describe the local scattering characteristics of the target from different aspects. The complementary information can further improve recognition performance. In addition, the direct application of both algorithms results in many scattering topological points located in the background region being extracted incorrectly, which is harmful to build the topological graph of the target. Therefore, we design different postprocessing processes to remove the incorrectly extracted scattering topological points as much as possible. After obtaining the correctly extracted scattering topological points, the connectivity between different points is defined based on their similarity to construct the topological graph. The details are described as follows.

1) ASC-Based Scattering Topological Point Extraction: The ASC model [49] is widely used in the SAR ATR field. In the high-frequency region, the model considers that the radar backscattering of a target can be approximated as a sum of responses from individual scattering centers, which can be expressed as follows:

$$E(f, \varphi; \Theta) = \sum_{i=1}^{p} E_i(f, \varphi; \theta_i) \tag{4}$$

where $E(f, \varphi; \Theta)$ represents the total echo of p individual scattering centers; f represents the frequency; φ represents the aspect angle; and $\Theta = \{\theta_i\} = [A_i, \alpha_i, x_i, y_i, L_i, \overline{\varphi}_i, \gamma_i](i = 1, 2, \ldots, p)$ represents the attribute parameter set of scattering centers. For the ith scattering center, the backscattering field can be described as

$$\begin{aligned} E_i(f, \varphi; \theta_i) = A_i \cdot &\left(j\frac{f}{f_c}\right)^{\alpha_i} \\ &\cdot \exp\left(-j\frac{4\pi f}{c}(x_i \cos\varphi + y_i \sin\varphi)\right) \\ &\cdot \text{sinc}\left(\frac{2\pi f}{c} L_i \sin(\varphi - \overline{\varphi}_i)\right) \\ &\cdot \exp(-2\pi f \gamma_i \sin\varphi) \end{aligned} \tag{5}$$

where $\text{sinc}(\cdot) = \sin(\cdot)/(\cdot)$, f_c is the radar center frequency, and c denotes the propagation velocity of the electromagnetic

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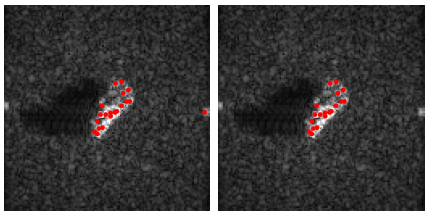


Fig. 6. Visualization of ASCs. (a) All ASCs. (b) ASCs after screening.

wave. In the attributes of the ith scattering center, A_i denotes the complex magnitude; α_i denotes the frequency dependence; x_i and y_i represent the position coordinates in the range dimension and azimuth dimension, respectively; L_i and $\overline{\varphi}_i$ are the length and orientation of the distributed scattering center; and γ_i is the aspect dependence of the localized scattering center. In addition, the scattering centers of the target are mainly located at the discontinuous parts of the target, such as edges, corners, and tips, representing the fine physical structure of the target, and thus, the ASC model is suitable for scattering topological points' extraction. At present, there are many optimization algorithms that can estimate these seven parameters. In this article, the sparse representation algorithm proposed in [20] is used for ASC extraction.

According to Zhang et al. [28], when the ratio of the radar system bandwidth to the center frequency is small, the frequency-dependent factor α_i could be ignored. Meanwhile, γ_i is usually small for a SAR system. The complex magnitude A_i can be decomposed into modulus $|A_i|$ and phase ψ_i, where ψ_i falls into $[0, 2\pi]$. Thus, the ith individual scattering center can be represented by an attribute feature vector $\mathbf{v}_{\text{ASC},i}$, where $\mathbf{v}_{\text{ASC},i} = [|A_i|, \psi_i, L_i, \overline{\varphi}_i, x_i, y_i] \in \mathbb{R}^6$.

It is important to note that, when using the sparse representation method proposed in [20] for ASC extraction, the number of scattering centers extracted from each image is the same (i.e., p). Unfortunately, a fixed number of scattering centers is not suitable for all images. As shown in Fig. 6(a), the scattering centers located in the background area may be extracted in some images, which is harmful to the construction of the scattering topology. To solve the problem, we remove the incorrectly extracted scattering centers by coordinates before constructing the graph. Concretely, we first calculate the barycenter $(\overline{x}, \overline{y})$ of all scattering centers, which can be denoted as (6) and (7). Then, the Euclidean distance between each scattering center (x_i, y_i) and the barycenter $(\overline{x}, \overline{y})$ is calculated. After that, a threshold τ_{dis} is used to remove scattering centers that are too far from the barycenter. The correctly extracted scattering centers are shown in Fig. 6(b)

$$\overline{x} = \frac{\sum_{i=1}^{p} |A_i| \cdot x_i}{\sum_{i=1}^{p} |A_i|} \tag{6}$$

$$\overline{y} = \frac{\sum_{i=1}^{p} |A_i| \cdot y_i}{\sum_{i=1}^{p} |A_i|}. \tag{7}$$

Suppose there are $q_{\text{ASC}} (q_{\text{ASC}} \leq p)$ attribute feature vectors left, where q_{ASC} is not fixed in different images. Thus, the target can be represented by a set of attribute feature vectors $\mathcal{V}_{\text{ASC}}$, where $\mathcal{V}_{\text{ASC}} = \{\mathbf{v}_{\text{ASC},i}\}_{i=1}^{q_{\text{ASC}}}$.

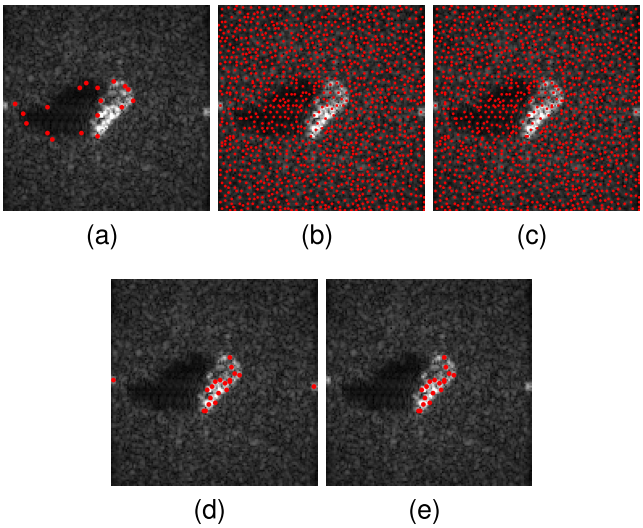


Fig. 7. Visualization of keypoints. (a) Corner points. (b) Strong scattering points. (c) All keypoints. (d) Keypoints after the first screening. (e) Keypoints after the second screening. Since each keypoint is drawn relatively large for visualization, there are some overlaps.

2) SAR-SIFT-Based Scattering Topological Point Extraction: The SIFT algorithm [50] is one of the most classic keypoint detection and local feature description algorithms. Its invariances to scale changes, rotations, translations, and partial to illumination changes and affine distortions make it widely used in the field of computer vision. However, the original SIFT is proposed for optical images and does not perform well on SAR images containing multiplicative speckle noise. The SAR-SIFT algorithm is an improved version of SIFT for SAR images, mitigating the effect of speckle noise on gradient calculation and feature extraction. Therefore, the SAR-SIFT algorithm is selected in this article to extract keypoints from SAR images. Like the SIFT algorithm, the SAR-SIFT algorithm not only extracts the keypoints of the image but also generates a descriptor for each keypoint to describe its neighborhood gradient information. The calculation process of SAR-SIFT can be divided into three steps: keypoint detection, orientation assignment, and descriptor extraction. The details are described in [21].

In fact, keypoint detection algorithms are mostly used for the image registration task [73], [74], while this article focuses on the target recognition task. The keypoints in the image registration task are the same in nature as the scattering topological points in this article, being important or unique elements in the image and having some invariance to different observation conditions. However, there are some differences between the keypoints and the scattering topological points due to the specific tasks. In image registration, more attention is paid to the consistency of the local structure on different images, while, in this article, more attention is paid to the topological structure of the whole target.

Generally, the keypoints extracted by the SAR-SIFT algorithm are corner points. Since the vehicle target has a complex geometry, corner points can portray its topology to some extent. However, as shown in Fig. 7(a), the number of corner points is often small. Therefore, we additionally introduce strong scattering points extracted based on the

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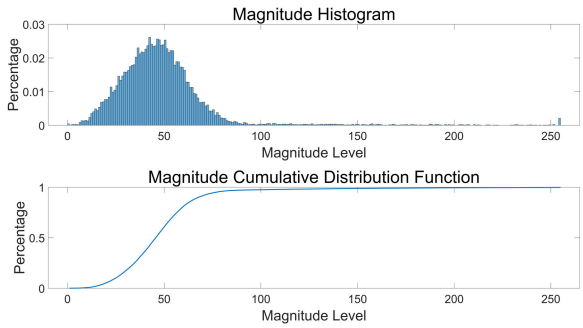


Fig. 8. Illustration of magnitude histogram and corresponding cdf.

clustering algorithm [8], [75] into the keypoint detection step to fully describe the scattering topology of the target. Thus, the final scattering topological points of the target are composed of corner points and strong scattering points together. The extracted strong scattering points and the final scattering topological points are shown in Fig. 7(b) and (c), respectively.

Since the processes of extracting corner points and strong scattering points are both performed on the whole image, many keypoints located in the background area will also be extracted. Thus, all the extracted keypoints are screened twice to remove the incorrectly extracted points as much as possible. The first screening is based on pixel magnitudes. Specifically, we first count the magnitude histogram of the SAR image and then accumulate the histogram to obtain the magnitude cumulative distribution function (cdf) that can completely describe the probability distribution of magnitude. Fig. 8 shows the magnitude histogram and the corresponding cdf of the SAR image in Fig. 7. The purpose of calculating the cdf is to adaptively determine the magnitude threshold for each image. We regard the keypoints whose magnitudes lie within the maximum 1% of the cdf as the scattering topological points on the target. Considering that the cdf is discrete, the threshold is set as the minimum of the horizontal coordinates of all points on the cdf whose vertical coordinates are greater than or equal to 0.99. Then, all keypoints with magnitudes less than the threshold are removed. Compared with simply setting a global magnitude threshold, the threshold calculated from the cdf is more flexible. The second screening is based on the coordinates and is the same as the method used in Section III-C1. First, the barycenter of the remaining keypoints is calculated. Then, the keypoints whose distances from the barycenter are larger than the threshold τ_{dis} are removed. The keypoints after the first screening and the second screening are shown in Fig. 7(d) and (e), respectively.

After obtaining the correctly extracted scattering topological points, we proceed to the second and third steps of the original SAR-SIFT algorithm to assign a principal orientation to each scattering topological point and generate the corresponding descriptor. After the whole process, the target can be represented as the SAR-SIFT descriptors corresponding to its scattering topological points. Suppose that there are q_{SST} scattering topological points left after two screenings. The target can be represented by a set of SAR-SIFT descriptors $\mathcal{V}_{\text{SST}}$, where $\mathcal{V}_{\text{SST}} = \{\mathbf{v}_{\text{SST},i}\}_{i=1}^{q_{\text{SST}}}$ and $\mathbf{v}_{\text{SST},i}$ is the SAR-SIFT descriptor corresponding to the ith scattering topological point.

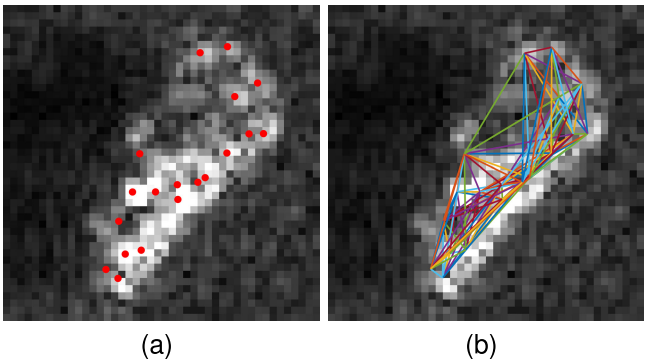


Fig. 9. Visualization of scattering topological graph. (a) Scattering topological points. (b) Corresponding topological graph. For clearer visualizations, different colors are set to different edges.

Since the descriptor is obtained by sequentially concatenating the 12-D gradient orientation histograms of nine subregions centered on the scattering topological point, its dimensionality is 108. Note that the number q_{SST} may vary in different images. The numbers q_{ASC} and q_{SST} may also be different for the same image.

3) Graph Structure Construction: After extracting scattering topological points, the undirected unweighted graph is constructed for each image to model its scattering topology. Graph data are mainly composed of the set of nodes and the set of edges. Generally, the adjacency matrix $\mathbf{M}$ is used to store edge information, where $m_{i,j} = 1$ means that there is an edge from the ith node to the jth node.

Suppose that $\mathcal{V} = \{\mathbf{v}_i\}_{i=1}^{q}$ denotes an arbitrary set with q scattering topological points. In fact, these scattering topological points can be viewed as a set of unordered nodes. According to the cosine similarity between different nodes, the adjacency matrix is defined to describe the connectivity, which is formulated as

$$m_{i,j} = \begin{cases} 1, & \text{if } \dfrac{\langle \mathbf{v}_i, \mathbf{v}_j \rangle}{\|\mathbf{v}_i\| \cdot \|\mathbf{v}_j\|} \geq \tau_{\text{sim}} \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product of vectors and τ_{sim} is the similarity threshold used to filter noisy links. After that, we obtain the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ that contains the local scattering information and spatial topology of the target, where $\mathcal{E}$ denotes all the edges. A set of scattering topological points and the corresponding topological graph are shown in Fig. 9.

D. Scattering Topological Feature Extraction Module

Benefiting from the mechanism of message passing, GNNs can integrate spatial topological information into node feature learning. Therefore, GNNs are introduced in this article to extract the scattering topological features. Besides, GNNs have the ability to extract features of the same dimension from graphs with different structures, which greatly facilitates the subsequent processing.

Concretely speaking, two GNNs, namely, GNN_{ASC} and GNN_{SST}, are designed to extract features from $\mathcal{G}_{\text{ASC}}$ and $\mathcal{G}_{\text{SST}}$, respectively. In fact, the architectures of GNN_{ASC} and GNN_{SST} are identical. However, the dimensionalities of the

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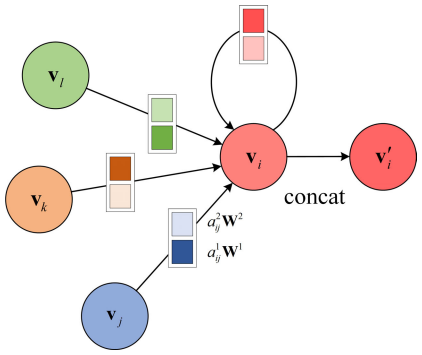


Fig. 10. Illustration of node feature update with multihead attention (with $K = 2$) graph convolutional layer. a is the attention weight, and $\mathbf{W}$ is the linear transformation matrix.

feature transformation matrices in their first layer are different, resulting in GNN_{ASC} only being able to extract features from $\mathcal{G}_{\text{ASC}}$ and GNN_{SST} only being able to extract features from $\mathcal{G}_{\text{SST}}$. Mathematically, for any graph $\mathcal{G}$, the GNN can be parameterized as the following equation:

$$\mathbf{z}_{\text{st}} = \text{GNN}(\mathcal{G}; \boldsymbol{\omega}_{\text{st}}) \tag{9}$$

where $\boldsymbol{\omega}_{\text{st}}$ and $\text{GNN}(\cdot)$ denote the parameters and the nonlinear mapping process of the GNN, respectively. $\mathbf{z}_{\text{st}}$ represents the extracted scattering topological features. The subscript $(\cdot)_{\text{st}}$ will be replaced by $(\cdot)_{\text{ASC}}$ or $(\cdot)_{\text{SST}}$ when we need to distinguish GNN_{ASC} and GNN_{SST} in the following.

In this article, the graph attention layer [42] is adopted to build the GNN. The graph attentional layer takes both node features $\mathcal{V}$ and edges $\mathcal{E}$ as input. Benefitting from the attention mechanism [51], this layer can selectively aggregate information from connected nodes, which improves the ability to represent complex graph structures.

Mathematically, for the feature $\mathbf{v}_i \in \mathbb{R}^F$ of the ith node, the process of node feature aggregation can be denoted as

$$\mathbf{v}'_i = \Big\|_{k=1}^{K} \sigma\left(\sum_{j \in \mathcal{N}_i} a^k_{ij} \mathbf{W}^k \mathbf{v}_i\right) \tag{10}$$

where $\mathbf{v}'_i$ is the aggregated node feature. $\|$ is the concatenation operation, and K is the number of attention heads. $\sigma(\cdot)$ represents the nonlinear activation function. According to Veli et al. [42], the exponential linear unit (ELU) activation function [48] is used here. $\mathcal{N}_i$ represents the set of nodes that have edges with the ith node (including i, i.e., the self-loop of each node is allowed). In the kth attention head, $\mathbf{W}^k \in \mathbb{R}^{F' \times F}$ is a learnable weight matrix shared among nodes for linear transformation of each node. a^k_{ij} is the normalized attention coefficient between the ith node and its jth neighbor node, which is calculated as

$$a^k_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{w}^{k^T}_{\text{att}}\left[\mathbf{W}^k \mathbf{v}_i \| \mathbf{W}^k \mathbf{v}_j\right]\right)\right)}{\sum_{l \in \mathcal{N}_i} \exp\left(\text{LeakyReLU}\left(\mathbf{w}^{k^T}_{\text{att}}\left[\mathbf{W}^k \mathbf{v}_i \| \mathbf{W}^k \mathbf{v}_l\right]\right)\right)} \tag{11}$$

where $(\cdot)^T$ represents transposition and $\mathbf{w}^k_{\text{att}} \in \mathbb{R}^{2F'}$ is a learnable weight vector shared among nodes to calculate the correlation between two nodes. Fig. 10 illustrates the node feature update process of a node with three neighbor nodes under two-head attention.

Similar to stacking multiple convolutional layers to build a CNN, the GNN is built by stacking multiple graph attention layers. During the optimization of GNN, the node features are easily oversmoothed as the network goes deeper, so we only stacked two graph attention layers. Specifically, the first layer and the second layer both consist of $K = 4$ attention heads. Each head maps the node features into $F' = 64$ dimensions. Therefore, the total output features have a dimension of 256. Then, the elementwise max pooling operation is performed to aggregate information from all node features to generate a global scattering topological feature that contains the scattering characteristics and spatial topology.

E. Feature Fusion and Classification Module

After obtaining the visual feature $\mathbf{z}_{\text{vis}} \in \mathbb{R}^{512}$ and the scattering topological features $\mathbf{z}_{\text{ASC}} \in \mathbb{R}^{256}$ and $\mathbf{z}_{\text{SST}} \in \mathbb{R}^{256}$, we concatenate them to generate a more effective and generalized feature $\mathbf{z}_{\text{cat}} \in \mathbb{R}^{1024}$, which can be described as

$$\mathbf{z}_{\text{cat}} = [\mathbf{z}_{\text{vis}} \| \mathbf{z}_{\text{ASC}} \| \mathbf{z}_{\text{SST}}]. \tag{12}$$

Then, the FC layer and the ReLU activation function are used to recode $\mathbf{z}_{cat}$ to remove the redundant information, which can be parameterized as

$$\mathbf{z}_{\text{fuse}} = \text{FC}_{\text{fuse}}(\mathbf{z}_{\text{cat}}; \boldsymbol{\omega}_{\text{fuse}}) \tag{13}$$

where $\boldsymbol{\omega}_{\text{fuse}}$ and $\text{FC}_{\text{fuse}}(\cdot)$ denote the parameters and the nonlinear mapping process, respectively. $\mathbf{z}_{\text{fuse}} \in \mathbb{R}^d$ denotes the output fused feature, and d is set to 128 in this article.

Subsequently, another FC layer and the softmax function are used to predict the label, which can be described as

$$\bar{\mathbf{y}} = \text{FC}_{\text{cls}}(\mathbf{z}_{\text{fuse}}; \boldsymbol{\omega}_{\text{cls}}) \tag{14}$$

where $\boldsymbol{\omega}_{\text{cls}}$ and $\text{FC}_{\text{cls}}(\cdot)$ denote the parameters and the nonlinear mapping process, respectively. $\bar{\mathbf{y}}$ is the predicted label of sample $\mathbf{X}$.

The cross-entropy function is used as the classification loss, which can be expressed as

$$\text{Loss}_{\text{cls}} = -\frac{1}{n^s} \sum_{i=1}^{n^s} \left\langle \mathbf{y}^s_i, \log\left(\bar{\mathbf{y}}^s_i\right) \right\rangle \tag{15}$$

where $\mathbf{y}^s_i$ is a one-hot vector indicating the label of the ith source domain sample $\mathbf{X}^s_i$ and $\bar{\mathbf{y}}^s_i$ is the corresponding predicted label.

In this article, the fused feature $\mathbf{z}_{\text{fuse}}$ is used to calculate the domain adaptation loss. Thus, the source domain $\mathcal{D}^s$ and the target domain $\mathcal{D}^t$ can be represented by $\mathbf{Z}^s_{\text{fuse}} = [\mathbf{z}^s_{\text{fuse},1}{}^T; \ldots; \mathbf{z}^s_{\text{fuse},n^s}{}^T] \in \mathbb{R}^{n^s \times d}$ and $\mathbf{Z}^t_{\text{fuse}} = [\mathbf{z}^t_{\text{fuse},1}{}^T; \ldots; \mathbf{z}^t_{\text{fuse},n^t}{}^T] \in \mathbb{R}^{n^t \times d}$. d is the dimension of the fused feature $\mathbf{z}_{\text{fuse}}$ and is equal to 128. Earlier studies propose several global domain adaptation losses [23], [24]. The correlation alignment (CORAL) loss [24] is used in this article, which measures the correlation between different features in the source and target domains and is defined as

$$\text{CORAL}\left(\mathcal{D}^s, \mathcal{D}^t\right) = \frac{1}{4d^2} \left\|\mathbf{C}^s - \mathbf{C}^t\right\|^2_F \tag{16}$$

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where $\mathbf{C}^s$ and $\mathbf{C}^t$ represent the correlation metrics of $\mathbf{Z}^s_{\text{fuse}}$ and $\mathbf{Z}^t_{\text{fuse}}$. Sun and Saenko [24] use the covariance matrix to measure the correlation between different features, i.e., $\mathbf{C}$ is defined as

$$\mathbf{C} = \frac{1}{n-1}\left(\mathbf{Z}^T_{\text{fuse}}\mathbf{Z}_{\text{fuse}} - \frac{1}{n}\left(\mathbf{1}^T\mathbf{Z}_{\text{fuse}}\right)^T\left(\mathbf{1}^T\mathbf{Z}_{\text{fuse}}\right)\right) \tag{17}$$

where $\mathbf{1}$ is a column vector with all elements equal to 1 and n is the number of samples in $\mathbf{Z}_{\text{fuse}}$. In this article, the subtracted mean term [i.e., the second term on the right-hand side of (17)] is ignored, and we directly utilize the inner product matrix to measure the correlation between different features, i.e., $\mathbf{C}$ is defined as

$$\mathbf{C} = \frac{1}{n-1}\left(\mathbf{Z}^T_{\text{fuse}}\mathbf{Z}_{\text{fuse}}\right). \tag{18}$$

Since the mean is not subtracted from the features, aligning the inner product matrices between the source and target domains is more difficult than aligning the covariance matrices, thus providing a stronger constraint and improving the transfer performance. Unfortunately, the global domain adaptation loss cannot control the distribution of subdomains because only the statistical distribution between the whole source domain and the whole target domain is constrained. To address this problem, some researchers propose a local domain adaptation loss named local maximum mean discrepancy (LMMD) [22] to assign the target domain samples to different subdomains by prediction confidences and align the distribution of source and target domains class by class, which is defined as

$$\begin{aligned}
&\text{LMMD}\left(\mathcal{D}^s, \mathcal{D}^t\right) \\
&= \frac{1}{L}\sum_{l=1}^{L}\left\|\sum_{i=1}^{n^s}\omega_i^{sl}\phi\left(\mathbf{z}^s_{\text{fuse},i}\right) - \sum_{j=1}^{n^t}\omega_j^{tl}\phi\left(\mathbf{z}^t_{\text{fuse},j}\right)\right\|^2_{\mathcal{H}} \\
&= \frac{1}{L}\sum_{l=1}^{L}\left[\sum_{i=1}^{n^s}\sum_{j=1}^{n^s}\omega_i^{sl}\omega_j^{sl}k\left(\mathbf{z}^s_{\text{fuse},i}, \mathbf{z}^s_{\text{fuse},j}\right)\right. \\
&\qquad + \sum_{i=1}^{n^t}\sum_{j=1}^{n^t}\omega_i^{tl}\omega_j^{tl}k\left(\mathbf{z}^t_{\text{fuse},i}, \mathbf{z}^t_{\text{fuse},j}\right) \\
&\qquad \left. -2\sum_{i=1}^{n^s}\sum_{j=1}^{n^t}\omega_i^{sl}\omega_j^{tl}k\left(\mathbf{z}^s_{\text{fuse},i}, \mathbf{z}^t_{\text{fuse},j}\right)\right]
\end{aligned} \tag{19}$$

where L is the number of classes and $\mathcal{H}$ is the reproducing kernel Hilbert space (RKHS) endowed with a kernel function k. $\phi(\cdot)$ denotes the feature mapping to map fused features to RKHS, and $k(\mathbf{z}^s_{\text{fuse},i}, \mathbf{z}^t_{\text{fuse},j}) = \langle\phi(\mathbf{z}^s_{\text{fuse},i}), \phi(\mathbf{z}^t_{\text{fuse},j})\rangle$. Consistent with [22], we adopt a Gaussian kernel with bandwidth set to median pairwise squared distances of any two samples in the source and target domains [52]. ω_i^{sl} and ω_i^{tl} denote the weights of $\mathbf{z}^s_{\text{fuse},i}$ and $\mathbf{z}^t_{\text{fuse},i}$ belonging to class l, respectively. The calculation processes are shown as follows:

$$\omega_i^{sl} = \frac{y_i^{sl}}{\sum_{j,\left(\mathbf{X}^s_j, \mathbf{y}^s_j\right)\in\mathcal{D}^s} y_j^{sl}} \tag{20}$$

$$\omega_i^{tl} = \frac{\overline{y}_i^{tl}}{\sum_{j,\left(\mathbf{X}^t_j, \overline{\mathbf{y}}^t_j\right)\in\mathcal{D}^t} \overline{y}_j^{tl}} \tag{21}$$

Algorithm 1 Training Process of the Proposed VSFA

Input: Source domain $\mathcal{D}^s = \{(\mathbf{X}^s_i, \mathbf{y}^s_i)\}_{i=1}^{n^s}$, target domain $\mathcal{D}^t = \{\mathbf{X}^t_j\}_{j=1}^{n^t}$.
Output: Model parameter set $\Omega = \{\boldsymbol{\omega}_{\text{en}}, \boldsymbol{\omega}_{\text{de}}, \boldsymbol{\omega}_{\text{ASC}}, \boldsymbol{\omega}_{\text{SST}}, \boldsymbol{\omega}_{\text{fuse}}, \boldsymbol{\omega}_{\text{cls}}\}$
for epoch $= 1, 2, \ldots,$ **do**
1: Extract visual features $\mathbf{z}_{\text{vis}}$ from $\mathbf{X}$ according to (1);
2: Reconstruct the images from visual features to get $\hat{\mathbf{X}}$ with (2);
3: Extract scattering topological point sets $\mathcal{V}_{\text{ASC}}$ based on ASC model;
4: Extract scattering topological point sets $\mathcal{V}_{\text{SST}}$ based on SAR-SIFT algorithm;
5: Construct two graphs $\mathcal{G}_{\text{ASC}}$ and $\mathcal{G}_{\text{SST}}$ according to (8);
6: Extract scattering topological features $\mathbf{z}_{\text{ASC}}$ and $\mathbf{z}_{\text{SST}}$ from corresponding graphs according to (9);
7: Concatenate the visual features and two scattering topological features to obtain $\mathbf{z}_{\text{cat}}$ with (12);
8: Recode $\mathbf{z}_{\text{cat}}$ to obtain the fused features $\mathbf{z}_{\text{fuse}}$ in accordance with (13);
9: Predict the label $\overline{\mathbf{y}}$ according to (14);
10: Compute the reconstruction loss Loss_{rec}, classification loss Loss_{cls}, and domain adaptation loss Loss_{da} according to (3), (15), and (22), respectively;
10: Obtain the total loss $\text{Loss}_{\text{total}}$ with (23);
10: Update the model parameter set Ω via backward propagation;
end for

where y_i^{sl} is the lth element of the true one-hot label vector $\mathbf{y}^s_i$ of the source domain sample $\mathbf{X}^s_i$ and $\overline{y}_i^{tl}$ is the lth element of the predicted label vector $\overline{\mathbf{y}}^t_i$ of the target domain sample $\mathbf{X}^t_i$.

It can be found that the LMMD loss essentially measures the distance between the means of the source domain $\mathcal{D}^s$ and the target domain $\mathcal{D}^t$ in RKHS class by class. Therefore, the number of categories in the target and source domains should be the same. If the label spaces of the source and target domains are inconsistent, the samples belonging to the private label space will have no corresponding categories to match, resulting in negative transfer [68].

Since the true label $\mathbf{y}^s_i$ is known, it can be used to exactly compute ω_i^{sl} for each source sample. Therefore, each source domain sample will only participate in the subdomain alignment process of its own category. However, for each target sample, the true label $\mathbf{y}^t_i$ is unavailable in the training process, and what we can use is only the predicted label $\overline{\mathbf{y}}^t_i$. Thus, the target domain sample will participate in the subdomain alignment processes of all classes with different weights. It is obvious that the misclassified target sample will be involved in the subdomain alignment process of noncorresponding classes with greater weights than its own class. Therefore, subdomains are aligned in biased directions, limiting cross-domain recognition performance. Biased directions cause some samples to be misclassified with higher confidence, thus creating a vicious circle. The corresponding analysis is shown in Section IV-D.

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In order to mitigate the effect of misclassified samples on subdomain alignment, we propose a two-stage loss by combining the global domain adaptation loss and the local domain adaptation loss. Concretely, in the early stage of training, the global domain adaptation loss is used to align the global distribution between the source and target domains. Despite its relatively limited performance, the maximum prediction confidence of misclassified samples is relatively small. Therefore, misclassified samples have less impact on subdomains, providing a good initialization for the local domain adaptation loss. In the later stage of training, the global loss is replaced by the local loss, which can finely align the distribution of the same category in different domains to further improve the accuracy. The two-stage domain adaptation loss is formulated as

$$\text{Loss}_{\text{da}} = \begin{cases} \text{Loss}_{\text{global}}, & \text{epoch} \ \leq \tau_{\text{ep}} \\ \text{Loss}_{\text{local}}, & \text{otherwise} \end{cases} \tag{22}$$

where τ_{ep} is the threshold of training iteration epochs. In particular, instead of calculating the predicted labels of the target samples only once, we recalculate the predicted labels after updating parameters in each iteration when the local loss is applied. The benefit of continuously updating the predicted labels is that the obtained predicted labels become more accurate as the training proceeds, which, in turn, promotes subdomain alignments. This positive feedback enables the model to align different subdomains more and more accurately. However, the negative impact of a few misclassified samples on subdomain alignment is also amplified in the process.

In summary, the total loss function can be described as

$$\text{Loss}_{\text{total}} = \lambda_1 \cdot \text{Loss}_{\text{cls}} + \lambda_2 \cdot \text{Loss}_{\text{rec}} + \lambda_3 \cdot \text{Loss}_{\text{da}} \tag{23}$$

where λ_1, λ_2, and λ_3 are the weights. The training process of VSFA is depicted in Algorithm 1.

IV. Experiments

A. Dataset Description

To verify the cross-domain recognition performance of the proposed VSFA, we adopt the public SAMPLE dataset [7] containing (measured, synthetic) paired SAR images as the benchmark dataset. In fact, the measured images are obtained directly from the MSTAR dataset. For the synthetic images, the researchers first created the elaborate CAD models of targets and then used the asymptotic ray-tracing techniques to generate the corresponding synthetic images [7]. During the simulation process, the target configurations and sensor parameters are matched to the measured data, which makes the SAMPLE dataset ideal for investigating the differences between synthetic and measured imageries. The magnitude statistics histogram of the synthetic and measured data is shown in Fig. 11, which verifies the distribution differences that exist between these two kinds of data again.

There are ten categories of ground vehicle targets in this dataset: 2S1, BMP2, BTR70, M1, M2, M35, M548, M60, T72, and ZSU23. Fig. 12 shows one pair of images of each category. These images are obtained by an X-band HH-polarization SAR with a resolution of 0.3 × 0.3 m and an image size of 128 × 128. The depression angle is from 14° to 17°. The range of azimuth angle is from 10° to 80°. The complete SAMPLE dataset has 1345 (measured, synthetic) paired SAR images. In each image pair, the synthetic image and the measured image are one-to-one correspondence, as reflected by the fact that their imaging parameters, such as the depression angle and the azimuth angle, are the same.

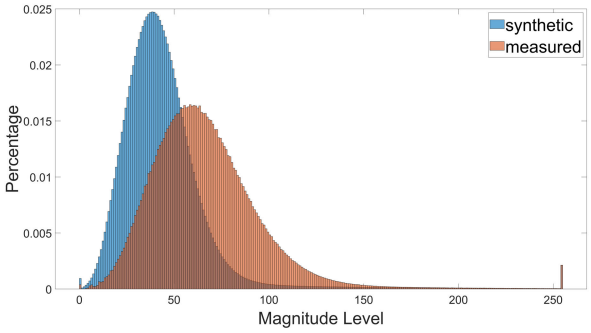


Fig. 11. Magnitude histogram of the synthetic and measured data.

At present, there are two mainstream experimental scenarios. In the first scenario, the training data consist of synthetic data with depression angles from 14° to 16°, while the test data are entirely composed of measured data with 17° depression angles. In other words, the training data and the test data are actually from different image pairs. Table I shows the specific number of targets in each category. It can be noticed that the number of synthetic and measured images in each category is entirely different, implying that there is no one-to-one correspondence between the synthetic and measured data in Scenario I. Therefore, this scenario is more in line with the actual situation. The second scenario is designed to evaluate the upper limit of the potential of the synthetic data. In this scenario, all the synthetic data are used for training, and all the measured data are regarded as the test data. The details are shown in Table II. It is obvious that, since the training and test data are paired, the second scenario has a lower recognition difficulty. For simplicity, these two scenarios are named **Scenario I** and **Scenario II**, respectively.

B. Experimental Setup

As mentioned in Section III-B, the input size of CNN is 64 × 64. The number of scattering topological points p extracted from each image by Yang et al. [20] is set to 20. To remove the incorrectly extracted scattering centers, the threshold τ_{dis} is set to 30, considering the size of ground vehicle targets. During constructing the graph, the threshold τ_{sim} used to remove noisy links is set to 0.0 and 0.6 for $\mathcal{G}_{\text{ASC}}$ and $\mathcal{G}_{\text{SST}}$, respectively. The VSFA is trained at a batch size of 256 for 150 epochs. The threshold τ_{ep} is set to 60. Moreover, the Adam algorithm [53] with an initial learning rate of 10^{-3} and a weight decay of 10^{-5} is used for the optimization. The weight coefficients λ_1, λ_2, and λ_3 are all set to 1 for simplicity. Our model is implemented using the PyTorch framework [54] and the Deep Graph Library (DGL) [55].

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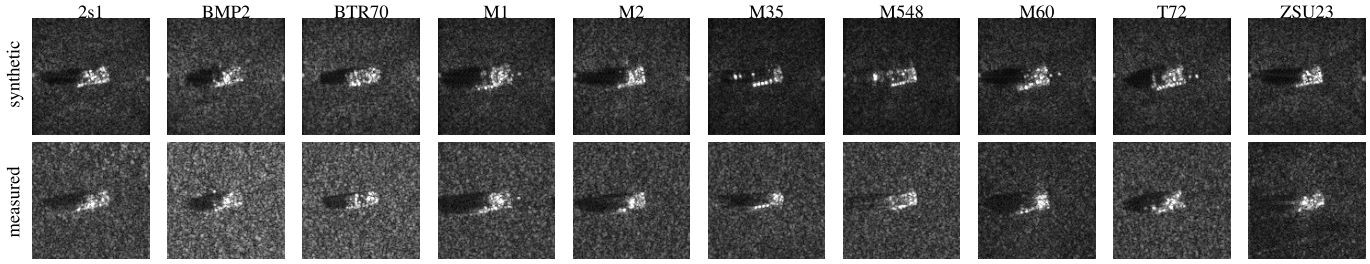


Fig. 12. Paired synthetic and measured images in the SAMPLE dataset.

TABLE I
DETAILS OF SCENARIO I

Class \ Depression Angle	Synthetic 14°∼16°	Measured 17°	Totals
2S1	116	58	174
BMP2	55	52	107
BTR70	43	49	92
M1	78	51	129
M2	75	53	128
M35	76	53	129
M548	75	53	128
M60	116	60	176
T72	56	52	108
ZSU23	116	58	174
Totals	806	539	1345

TABLE II
DETAILS OF SCENARIO II

Class \ Depression Angle	Synthetic 14°∼17°	Measured 14°∼17°	Totals
2S1	174	174	348
BMP2	107	107	214
BTR70	92	92	184
M1	129	129	258
M2	128	128	256
M35	129	129	258
M548	128	128	256
M60	176	176	352
T72	108	108	216
ZSU23	174	174	348
Totals	1345	1345	2690

C. Comparison Experiments

In order to fully verify the effectiveness of our method, we not only compare VSFA with recent algorithms proposed for the SAMPLE dataset [7], [16], [56], [57], [58], [59], [60], [61] but also compare VSFA with advanced unsupervised domain adaptation algorithms designed for optical images [63], [64], [65]. The performances of the former are directly quoted from the results reported in their original papers, while the performances of the latter are implemented according to their source codes. The results in the two experimental scenarios are shown in Tables III and IV, respectively. In particular, the results of VSFA are averaged over 50 replicate experiments. $\alpha \pm \beta\%$ means average $\pm$ standard deviation for the classification accuracy. Note that some existing methods do not report the standard deviations of the evaluation results. Since these authors do not publish their source codes, it is difficult to recover the standard deviations of the existing methods.

TABLE III
COMPARISON EXPERIMENTS IN SCENARIO I

Methods	Accuracy
Lewis (2019) [7]	24.97%
Scarnati (2019) [56]	55.62%
Lewis (2020) [57]	$82.05 \pm 0.53\%$
Li (2020) [64]	$88.57 \pm 1.33\%$
Li (2021) [65]	$91.28 \pm 2.58\%$
Inkawhich (2021) [16]	$95.06 \pm 0.87\%$
Wang (2019) [63]	$96.33 \pm 0.36\%$
Sun (2023) [61]	$97.36 \pm 0.87\%$
VSFA	$\mathbf{99.15 \pm 0.86}\%$

TABLE IV
COMPARISON EXPERIMENTS IN SCENARIO II

Methods	Accuracy
Li (2021) [65]	$89.62 \pm 2.24\%$
Li (2020) [64]	$90.97 \pm 2.04\%$
Dong (2022) [59]	93.66%
Sun (2023) [61]	$96.46 \pm 1.14\%$
Melzer (2021) [60]	96.88%
Wang (2019) [63]	$96.95 \pm 0.41\%$
VSFA	$\mathbf{98.18 \pm 0.97}\%$

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TABLE V
ACCURACY UNDER DIFFERENT DEPRESSION ANGLES

Depression Angle	Scenario I	Scenario II
14°∼16°	-	$97.25 \pm 1.58\%$
17°	$\mathbf{99.15 \pm 0.86\%}$	$99.58 \pm 0.39\%$
Total	-	$\mathbf{98.18 \pm 0.97\%}$

As shown in Tables III and IV, our proposed VSFA achieves the state-of-the-art performance in both experimental scenarios. Since Jennison et al. [58] do not explicitly indicate their experimental scenario, their 88.45% accuracy has not been added to Tables III and IV but is given here for comparison. In Scenario I, the accuracy is improved by 1.79% compared to the current best algorithm [61]. Similar to our proposed method, the algorithm proposed in [61] is based on the unsupervised domain adaptation method. The authors propose a gradual domain adaptation recognition framework with a pseudolabel denoising method. With the help of relatively accurate pseudolabels, the algorithm can also align subdomains class by class. However, the differences in scattering topologies are ignored in [61]. In Scenario II, the current best recognition performance is achieved by conditional adversarial network [66] with the domain-specific batch normalization layer [63]. The algorithm also utilizes the predicted label information of the target domain samples, thus achieving relatively favorable performances. Unfortunately, since the algorithm is proposed for optical images, it also does not take into account scattering topological differences. Compared with [63], the performance is improved by 1.23%. In addition, VSFA also outperforms the neural architecture search-based algorithm [60]. In general, neural architecture searching requires significant computational resources and is quite time-consuming. The significantly improved recognition performances in these two experimental scenarios entirely demonstrate the effectiveness of our algorithm.

Note that there seems to be an abnormal phenomenon in that the accuracy of Scenario II is lower than that of Scenario I. This is inconsistent with the theory, which states that the recognition difficulty of Scenario I is greater than that of Scenario II. In fact, these two results are not directly comparable due to different test data. Table V shows in detail the accuracy of the test data with different depression angles under Scenario II. It can be found that the accuracy of Scenario II is higher than that of Scenario I on the test data with 17° depression angle, which proves that the classification difficulty of Scenario II is indeed lower due to the one-to-one correspondence between the training and test data. In addition, from Table V, we can also find a characteristic of the SAMPLE dataset that the accuracy of data with 17° depression angle is significantly higher than that of data with depression angle from 14° to 16°, which is the direct reason why the accuracy of Scenario I seems higher than that of Scenario II.

Fig. 13 shows in detail the classification performance of the proposed VSFA under Scenario I and Scenario II in the form of confusion matrices. The confusion matrix is a visualization

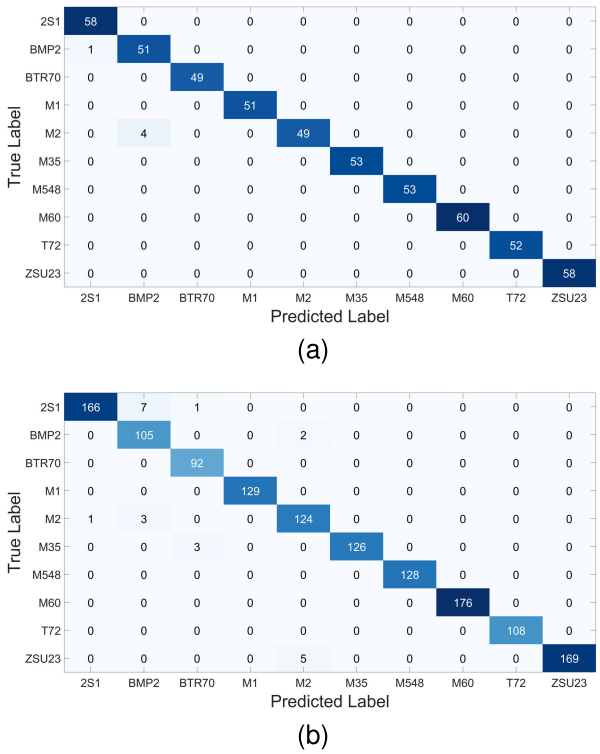


Fig. 13. Confusion matrices obtained by VSFA. (a) Scenario I. (b) Scenario II.

tool used to analyze the prediction results of the classification model, which summarizes the samples according to the real label and the predicted label in the form of the matrix. Its rows represent the real labels of samples, while columns correspond to the predicted labels of samples. The confusion matrices are calculated based on the predicted labels of the experiment closest to the average accuracy among 50 replicate experiments. As shown in Fig. 13, the values of the nondiagonal elements of the confusion matrices are very small for both Scenario I and Scenario II, which demonstrates that the proposed VSFA can achieve high recognition performance for each class of targets. In particular, the recognition rate of eight types of targets in Scenario I and five types of targets in Scenario II even reaches 100%.

D. Ablation Experiments

To evaluate the accuracy of the proposed method and analyze the influence of each component, this article conducts a series of ablation experiments on **Scenario I**.

1) Analysis on Each Component: The VSFA consists of five components: the encoder branch, the decoder branch (reconstruction loss), the GNN_{ASC} branch, the GNN_{SST} branch, and the proposed domain adaptation loss. It should be pointed out that the FC layers in the feature fusion and classification module are ignored here since they are necessary for classification. The impact of each component on performance is evaluated in two ways. On the one hand, the encoder branch is regarded as a baseline model to evaluate the contribution of other components to the performance. Specifically, we use the encoder branch to extract visual features from images and

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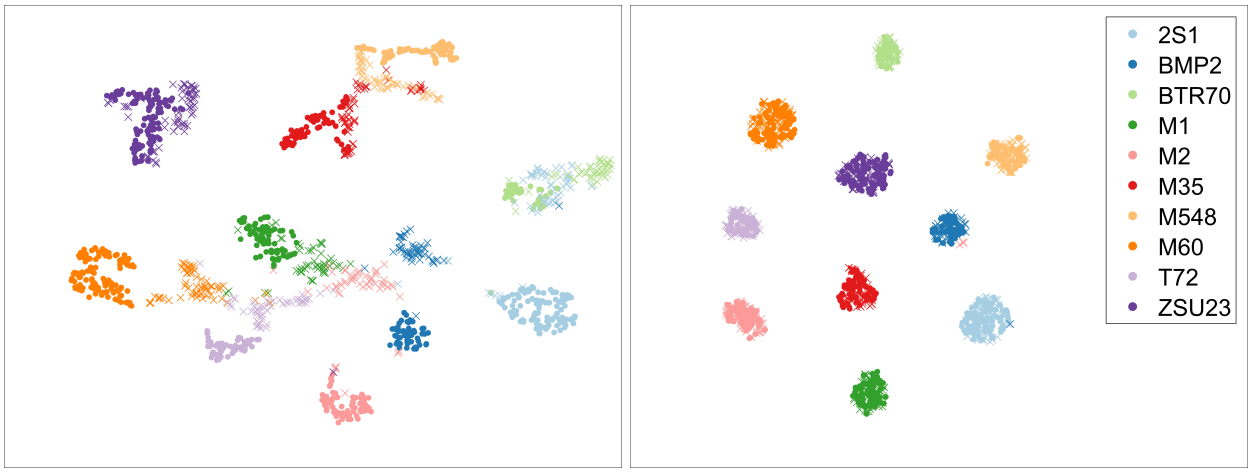


Fig. 14. Feature distribution visualization based on t-SNE embeddings. (a) Baseline. (b) VSFA. The same color indicates features extracted from samples of the same category in synthetic images (represented by ·) and measured images (represented by ×).

TABLE VI
ABLATION EXPERIMENT OF DIFFERENT COMPONENTS IN VSFA

Encoder	Decoder	GNN_{ASC}	GNN_{SST}	Loss_{da}	Accuracy
✓					$66.40 \pm 6.84\%$
✓	✓				$69.26 \pm 7.72\%$
✓	✓			✓	$97.23 \pm 1.13\%$
		✓			$56.34 \pm 1.97\%$
		✓		✓	$63.22 \pm 2.44\%$
			✓		$34.52 \pm 1.82\%$
			✓	✓	$42.24 \pm 2.83\%$
		✓	✓		$65.18 \pm 1.86\%$
		✓	✓	✓	$86.16 \pm 2.19\%$
✓	✓	✓		✓	$98.60 \pm 1.42\%$
✓	✓		✓	✓	$98.47 \pm 1.23\%$
✓	✓	✓	✓	✓	$\mathbf{99.15 \pm 0.86\%}$

feed these features into FC layers to predict their categories. Besides, only the cross-entropy classification loss is calculated in the baseline model. On the other hand, we evaluate the impact of scattering topological features on performance when visual features are not used. Similarly, we use the GNN_{ASC} branch or the GNN_{SST} branch to extract scattering topological features and feed them into FC layers to obtain their predicted labels. Besides, the convergence is relatively slow when using only GNNs to identify targets, and thus, we increase the number of training epochs to 300. The threshold τ_{ep} is also correspondingly increased to 120. The overall results of the ablation experiments are shown in Table VI.

As shown in Table VI, the recognition accuracy of the baseline model is only 66.40%, which also shows the huge difference between the synthetic and measured data. By introducing reconstruction as the auxiliary task, the accuracy can be increased by about 3%, which verifies that the generalization of visual features can be increased by simultaneously reconstructing source and target images. Without introducing any GNN branch, the recognition rate is improved to 97.23% using the proposed two-stage domain adaptation loss, demonstrating the effectiveness of domain adaptation loss in cross-domain recognition tasks. No matter which GNN branch is combined with the baseline model, the recognition accuracy can be improved, proving that it is beneficial to simultaneously align the distribution between the source and target domains in visual space and scattering topology space for cross-domain classification tasks. Since the ASC model and the SAR-SIFT algorithm reflect the local scattering characteristics of the target from different aspects, fusing the features extracted from $\mathcal{G}_{\text{ASC}}$ and $\mathcal{G}_{\text{SST}}$ can further improve the recognition rate. With the help of all components, the accuracy rate is finally improved to 99.15%. In addition, it can be found that the scattering topological features tend to be less effective than visual features under the same conditions. However, the performance is greatly improved after fusing two different scattering topological features. Without the domain adaptation loss, the recognition accuracy is improved to 65.18%, reaching the same level as visual features. After introducing the domain adaptation loss, the performance further reaches 86.16%. Although it is lower than the recognition rate of 97.23% obtained using visual features, the potential of scattering topological features is demonstrated to some extent.

The results of feature distribution visualization also demonstrate the effectiveness of VSFA. Concretely, the set of model parameters closest to the mean accuracy in 50 replicate experiments is selected to extract the fused features from images. Then, the t-SNE algorithm is utilized to embed 128-D fused features into a 2-D plane. As shown in Fig. 14, the whole VSFA achieves stronger intraclass compactness and interclass separability compared with the baseline model. In particular, the training images for each category are also more compactly distributed due to the simultaneous usage of visual information and scattering of topological information.

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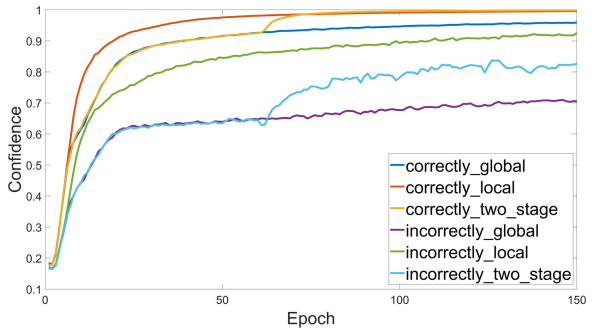


Fig. 15. Average confidence of correctly and incorrectly identified target domain data with different domain adaptation losses.

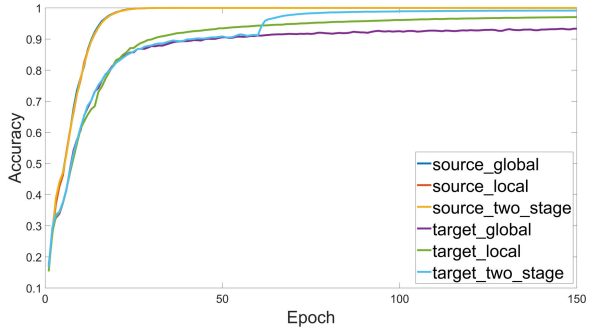


Fig. 16. Accuracy of source and target data with different domain adaptation losses.

TABLE VII
ACCURACY OF DIFFERENT DOMAIN ADAPTATION LOSSES

Scenario I	global loss	local loss	two-stage loss
Accuracy	$93.20 \pm 1.72\%$	$97.01 \pm 1.01\%$	$\mathbf{99.15 \pm 0.86}\%$

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT METRICS

Metric \ Loss	global loss	two-stage loss
Covariance	$92.42 \pm 2.18\%$	$98.58 \pm 1.29\%$
Inner Product	$93.20 \pm 1.72\%$	$99.15 \pm 0.86\%$

2) Analysis on Domain Adaptation Loss: In this section, we conduct a series of experiments to verify the validity of the proposed two-stage loss. Fig. 15 shows the variation of the average confidence of correctly and incorrectly classified target domain samples with different domain adaptation losses during the training process. With the progress of training, the classification decision boundary becomes clearer, and thus, the confidence of both correctly and incorrectly classified samples is rapidly increasing. The local domain adaptation loss aligns subdomains according to the prediction confidences of target domain samples. Therefore, misclassified target samples will participate in the subdomain alignment processes of noncorresponding classes with higher weights, limiting the correctness of subdomain alignments. Biased alignments cause some samples to be misclassified with higher confidence, thus creating a vicious circle.

In the early stage of training, the recognition ability of the network is relatively poor, resulting in many samples being misclassified. Although most misclassified samples can be corrected with the improvement of recognition ability, this vicious circle can still be observed in some samples. As shown in Fig. 15, the average confidence of misclassified samples under the constraint of the local loss increases significantly faster than that of misclassified samples under the constraint of the global loss. In particular, the average confidence of misclassified samples even reaches 0.9 in the later stage of the training process, meaning that they have almost the same impact on the subdomain corresponding to their predicted category as the samples that really belong to that category.

The global domain adaptation loss only aligns the overall distribution between the source and target domains, and does not affect the distribution between subdomains. Despite its relatively limited performance, the average confidence of misclassified samples is relatively low. Therefore, we leverage this characteristic to use the global domain adaptation loss in the early stage of the training process to improve the cross-domain recognition ability without significantly increasing the confidence of the misclassified samples. In the middle and later stages of the training process, the global loss is replaced by the local loss to align the distribution between subdomains accurately based on the preliminary alignment of the global distribution between the source and target domains, thus significantly improving the recognition accuracy. Although the average confidence still tends to increase as training proceeds, the average confidence of misclassified samples even decreases due to the improved recognition performance resulting from subdomain alignment when the global loss is just replaced with the local loss.

Fig. 16 shows the accuracy curves of source and target data under the constraint of different domain adaptation losses. It can be found that the source accuracy curves of these three kinds of domain adaptation loss are almost coincident, which means that they converge at the same rate. Because the local domain adaptation loss aligns the distribution between subdomains, its accuracy significantly exceeds the global loss. In the two-stage loss, there is a rapid and significant improvement in performance after the global loss is replaced by the local loss in the 60th epoch, which proves its effectiveness. The precise recognition accuracy under different domain adaptation losses is shown in Table VII. Compared to the local loss, our two-stage loss further significantly improves recognition performance by 2.14%.

Table VIII shows the performance comparison of using the inner product matrix and the covariance matrix in loss calculation in Scenario I. It can be found that the simple change of replacing the covariance matrix with the inner product matrix brings a performance gain of 0.78% when only the global loss is used. In the case of using the proposed two-stage loss combining the global loss and the local loss, the recognition accuracy is improved by 0.57%, which

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TABLE IX
ACCURACY WITH DIFFERENT τ_{ep}

τ_{ep}	0	10	20	30	40	50	60	70
Accuracy	$97.01 \pm 1.01\%$	$97.36 \pm 1.09\%$	$97.66 \pm 1.11\%$	$98.38 \pm 1.14\%$	$98.37 \pm 1.16\%$	$98.74 \pm 0.98\%$	$\mathbf{99.15 \pm 0.86}\%$	$99.09 \pm 0.88\%$
τ_{ep}	80	90	100	110	120	130	140	150
Accuracy	$99.13 \pm 0.94\%$	$99.14 \pm 0.88\%$	$99.03 \pm 1.18\%$	$99.09 \pm 0.96\%$	$98.94 \pm 1.05\%$	$98.54 \pm 1.31\%$	$97.76 \pm 1.47\%$	$93.20 \pm 1.72\%$

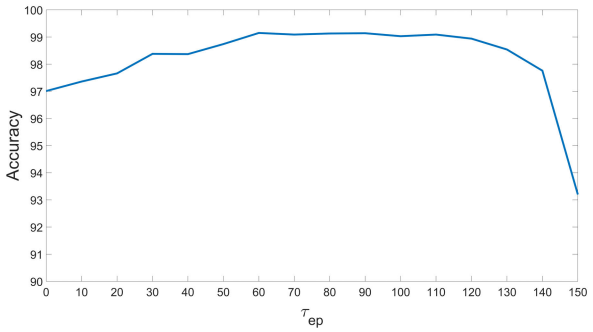


Fig. 17. Visualization of the influence of different τ_{ep}.

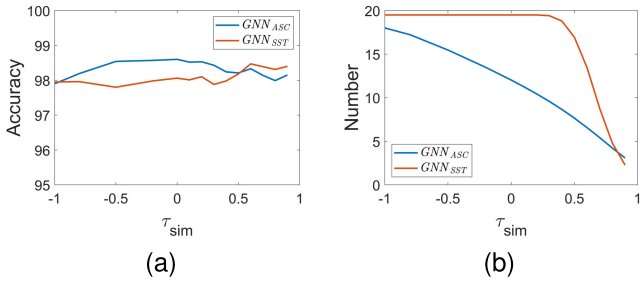


Fig. 18. Visualization of the influence of different τ_{sim}. (a) Recognition accuracy. (b) Average number of edges per node.

validates the effectiveness of the inner product matrix for loss calculation.

3) Analysis on Training Iteration Threshold: In the proposed two-stage domain adaptation loss, the training iteration threshold τ_{ep} is a hyperparameter used to control the number of epochs for the global and local losses participating in the network optimization and, therefore, affects the recognition performance. In order to explore the specific impact of τ_{ep} on recognition performance, we conduct a series of experiments in this section.

As shown in Fig. 17, the variation curve of recognition rate with threshold τ_{ep} can be approximated as a concave downward curve. Table IX shows the corresponding precise values in detail. It is worth noting that, when τ_{ep} equals 0 and 150, the proposed two-stage loss will degenerate into the local loss and the global loss, respectively. When the threshold τ_{ep} is too small, the local loss will replace the global loss prematurely. At this time, the performance of the network is relatively limited. Therefore, a large number of misclassified samples can have a negative impact on subclass alignment and, thus, reduce the recognition performance. On the contrary, if the threshold τ_{ep} is too large, the time for the local loss to participate in network optimization will be relatively short. Therefore, the local loss cannot align different subdomains well, thereby limiting the recognition performance. Fortunately, this problem can be easily solved by simply increasing the number of training epochs. For example, when τ_{ep} is set to 140, if the number of training epochs is increased from 150 to 200 (i.e., the number of epochs for the local loss participating in network optimization is increased from 10 to 60), the recognition performance will be improved from 97.76% to 99.04%. Furthermore, it can be found that the recognition performance is improved from 93.20% (when $\tau_{ep} = 150$ in Table IX) to 97.76% (when $\tau_{ep} = 140$ in Table IX) even with only 10 epochs of optimization using the local loss, indicating that subdomain alignment can quickly and significantly improve the recognition performance. Overall, we suggest replacing the global loss with the local loss after the performance of the network is relatively satisfactory. In addition, the local loss should be used for a sufficient number of iterations to accurately align the subdomains.

4) Analysis on Similarity Threshold: The similarity threshold τ_{sim} is a hyperparameter that controls the density of edges between nodes and, therefore, can influence the structure of the scattering topological graph. In this section, we conduct a series of experiments to investigate the effect of different hyperparameters on recognition performance. Note that, when exploring the influence of τ_{sim} on $\mathcal{G}_{ASC}$, we do not add the GNN_{SST} branch to the model and vice versa.

Fig. 18 visualizes the recognition performances and the average number of edges per node in $\mathcal{G}_{ASC}$ and $\mathcal{G}_{SST}$ under different threshold τ_{sim}. Table X shows the corresponding precise values in detail. Meanwhile, we statistically analyze the average similarity between all nodes in each graph and find that there is a significant difference between $\mathcal{G}_{ASC}$ and $\mathcal{G}_{SST}$, with an average similarity of 0.257 for the former and 0.645 for the latter. This is the reason why the average number of edges in $\mathcal{G}_{SST}$ is still 19.50 when the threshold τ_{sim} is increased to 0.2. In addition, this may be the reason why GNN_{ASC} is generally more effective than GNN_{SST} when only one GNN branch is added to the baseline. The higher average similarity implies that the different node features are more similar and, therefore, provide relatively limited information.

From Table X, we can see that our model is relatively robust to the threshold τ_{sim}. The difference between the lowest accuracy and the highest accuracy is only 0.7%. Moreover, the model will obtain relatively good performance when the threshold τ_{sim} is set near the average similarity. Too many edges will make the node features easily oversmoothed, while

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TABLE X
ACCURACY AND AVERAGE NUMBERS OF EDGES PER NODE WITH DIFFERENT τ_{sim}

τ_{sim}	$\mathcal{G}_{ASC}$		$\mathcal{G}_{SST}$	
	Accuracy	Number	Accuracy	Number
-1.0	$97.90 \pm 1.43\%$	18.01	$97.95 \pm 1.60\%$	19.50
-0.8	$98.19 \pm 1.30\%$	17.23	$97.96 \pm 1.16\%$	19.50
-0.5	$98.54 \pm 1.33\%$	15.47	$97.80 \pm 1.65\%$	19.50
-0.2	$98.57 \pm 1.22\%$	13.47	$97.98 \pm 1.42\%$	19.50
0.0	$\mathbf{98.60 \pm 1.42}\%$	12.02	$98.06 \pm 1.17\%$	19.50
0.1	$98.52 \pm 1.46\%$	11.25	$98.01 \pm 1.40\%$	19.50
0.2	$98.53 \pm 1.31\%$	10.44	$98.10 \pm 1.13\%$	19.50
0.3	$98.43 \pm 1.22\%$	9.58	$97.88 \pm 1.61\%$	19.42
0.4	$98.24 \pm 1.12\%$	8.65	$97.98 \pm 1.46\%$	18.81
0.5	$98.21 \pm 1.35\%$	7.65	$98.18 \pm 1.30\%$	16.93
0.6	$98.33 \pm 1.42\%$	6.57	$\mathbf{98.47 \pm 1.23}\%$	13.43
0.7	$98.14 \pm 1.57\%$	5.42	$98.39 \pm 1.08\%$	8.74
0.8	$97.99 \pm 1.60\%$	4.22	$98.31 \pm 1.10\%$	4.79
0.9	$98.15 \pm 1.59\%$	3.09	$98.40 \pm 1.10\%$	2.28

too few edges will make it difficult to exchange information between nodes. Therefore, the threshold τ_{sim} is finally set to 0.0 and 0.6 for $\mathcal{G}_{ASC}$ and $\mathcal{G}_{SST}$ in VSFA, respectively.

5) Analysis on the Number of Layers of GNNs: The stacking process of GNNs is different from that of CNNs. In general, the performance of a CNN will be generally improved as the network deepens. For example, ResNet has successfully extended the depth of CNNs to 152 layers and significantly improved the recognition performance. In contrast, the performance of a GNN usually degrades as the network goes deeper. Most GNNs are not deeper than four layers [67]. This is mainly due to the fact that most spatial-based GNNs are based on the message-passing mechanism, i.e., aggregating the information from the neighboring nodes of the central node and updating the central node based on the aggregated information and the information of the central node itself. As the network deepens, GNNs easily converge the features of all nodes to the same value, thus producing the oversmoothing problem, which significantly reduces recognition accuracy. Therefore, we evaluate the effect of different numbers of GNN layers on recognition performance in this section. Note that, unlike the experiments in Section IV-D4, the experiments in this subsection are performed on the full VSFA, i.e., the GNN_{ASC} branch and the GNN_{SST} branch are both added to the model.

As shown in Table XI, the performance of our model increases first and then decreases with the deepening of GNN_{ASC} and GNN_{SST}. When GNNs contain only one layer, their ability to learn the scattering topology of the target is insufficient, and therefore, the accuracy is relatively low. When GNNs have two or three layers, the network not only has strong learning ability but also is not oversmoothed,

TABLE XI
ACCURACY OF GNNS WITH DIFFERENT NUMBERS OF LAYERS

Layer	Accuracy
1	$98.29 \pm 1.20\%$
2	$\mathbf{99.15 \pm 0.86}\%$
3	$98.94 \pm 0.98\%$
4	$97.97 \pm 1.73\%$
5	$97.62 \pm 2.09\%$

TABLE XII
ANALYSIS OF TIME COST

Module	ASC-based graph construction	SAR-SIFT-based graph construction	Network	Total
Time	152.63s	89.59ms	6.12ms	152.73s

so it achieves relatively high performance. With the further deepening of GNNs, the oversmoothing problem is also aggravated, which seriously reduces the recognition performance of our model. Nevertheless, we can observe that the lowest recognition performance in Table XI is still higher than when the GNN_{ASC} branch and the GNN_{SST} branch are not added in VSFA, which indicates that, even if there is an oversmoothing problem, GNNs have learned some information that is beneficial to recognition tasks.

6) Analysis on Time Cost: In this section, we evaluate the time cost of the proposed algorithm to recognize a new image in the inference phase with a computer (CPU: Inter Xeon Silver 4210R 2.4 GHz; RAM: 128 GB; and GPU: RTX 3090). In order to model the scattering topology of the target, two different handcrafted scattering topological point extraction algorithms are introduced, which makes the proposed method not end-to-end. Therefore, we count the time costs of the ASC-based graph construction, the SAR-SIFT-based graph construction, and the forward propagation of the network, respectively.

As shown in Table XII, the proposed algorithm is not advantageous in terms of time cost. In fact, the inference speed of the network is relatively fast, while most of the time is spent on extracting scattering topological points and constructing the corresponding scattering topological graph. In particular, it accounts for more than 99.93% of the total time to construct the ASC-based scattering topological graph. It should be pointed out that the slow extraction speed is a common problem faced by current ASC extraction algorithms [20], [69], [70], [71], [72]. Despite its slow extraction speed, the ASC model has the excellent ability to describe the target's physical structure and local scattering characteristics. The seven parameters of the ASC model can represent the scattering center's amplitude, position, direction, length, and structure. For example, the frequency dependence factor α equal to 1 and the length L greater than 0 indicate that the structure here is a dihedral. The frequency dependence factor α equal to 0.5 and the length L equal to 0 indicate that the structure here

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is a top hat. Therefore, the ASC model is still consistently and widely used in SAR target recognition [27], [28], [29], [30], [31]. Inspired by these algorithms, we introduce the ASC model and explore how to model the scattering topology of the target by ASCs.

If faster ASC extraction algorithms are proposed in the future, our algorithm will also be accelerated. Besides, if the scattering topology of the target is modeled by the SAR-SIFT algorithm only, the total time will drop to about 95.71 ms. At this time, the accuracy of the proposed algorithm drops to 98.47% and 97.71% in the two experimental scenarios, respectively, which is still state-of-the-art compared to existing methods.

V. Conclusion

A novel unsupervised domain adaptation framework named VSFA is proposed in this article. This framework achieves accurate recognition of targets in measured images by synthetic images when the labels of the measured images are unknown and, thus, can be applied to the practical case where it is difficult to acquire the real labeled training target images. To the best of our knowledge, this is the first study to explore the differences in scattering topologies between synthetic and measured SAR images. Benefiting from simultaneously reducing the distribution differences between the synthetic and measured data in visual and scattering topological spaces, VSFA achieves state-of-the-art performance on the SAMPLE dataset. In addition, the ablation experiments also demonstrate the effectiveness of the components of the designed model, including the image reconstruction branch, two different but complementary scattering topological feature extraction branches, and the improved two-stage domain adaptation loss. Despite the promising results, the method currently has some room for improvement. For example, the extraction of scattering topological points is still based on some manually designed algorithms, which makes the whole framework not end-to-end and reduces the operational efficiency. Therefore, how to speed up the execution efficiency is a direction to be explored in the future. Nevertheless, we still believe that the simultaneous optimization of visual differences and scattering topological differences will have great potential in researching domain adaptation algorithms for SAR images. Our work can provide some clues for future research.

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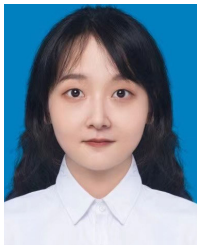
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